

Pharma Application Architecture

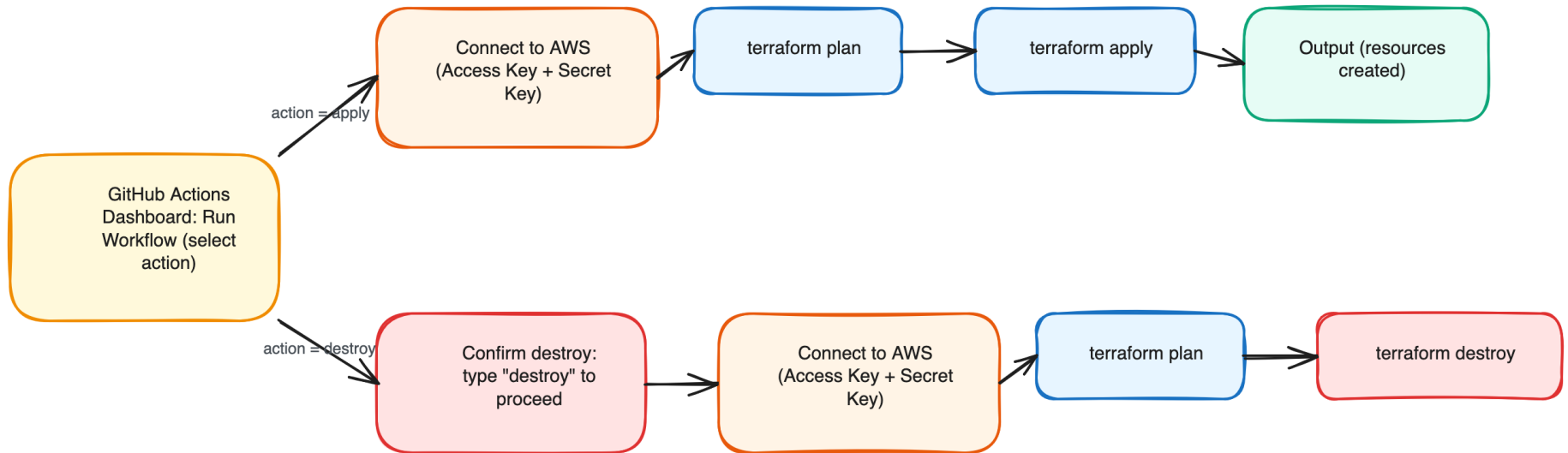
CI/CD, GitOps & Kubernetes Diagrams — zen-infra, frontend, backend, gitops

Generated August 02, 2026

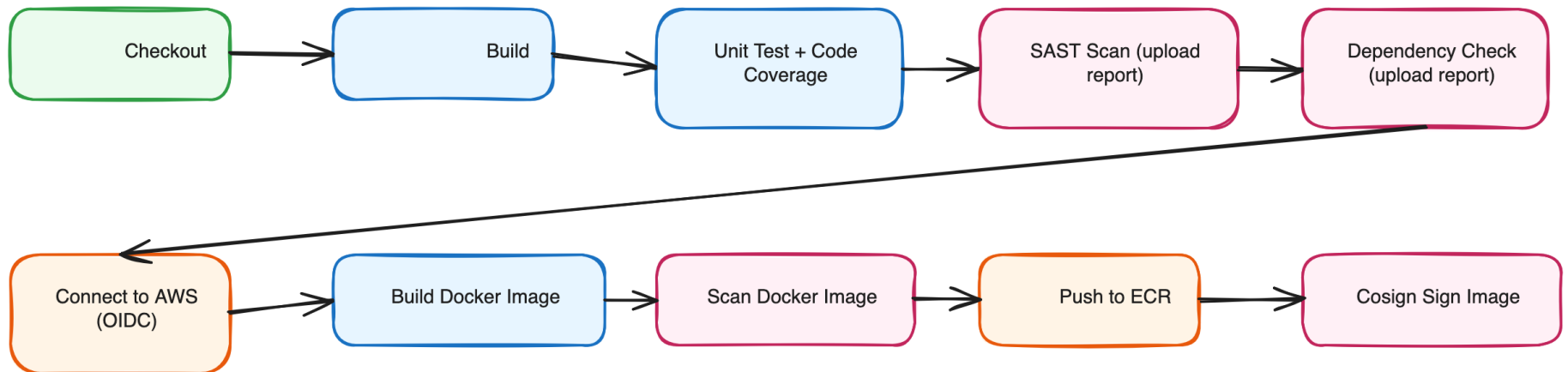
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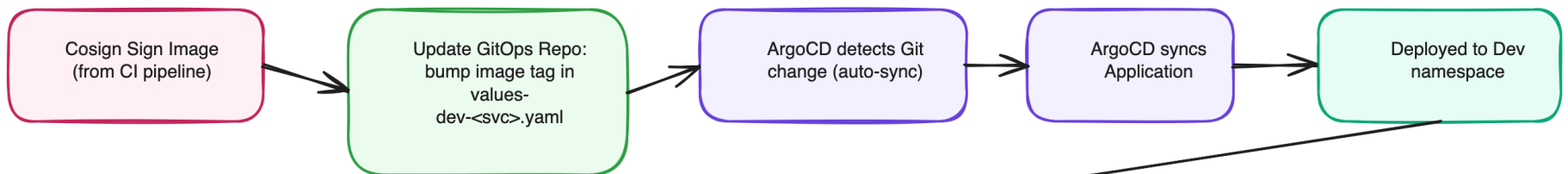
zen-infra: Terraform Apply & Destroy Workflow (GitHub Actions)



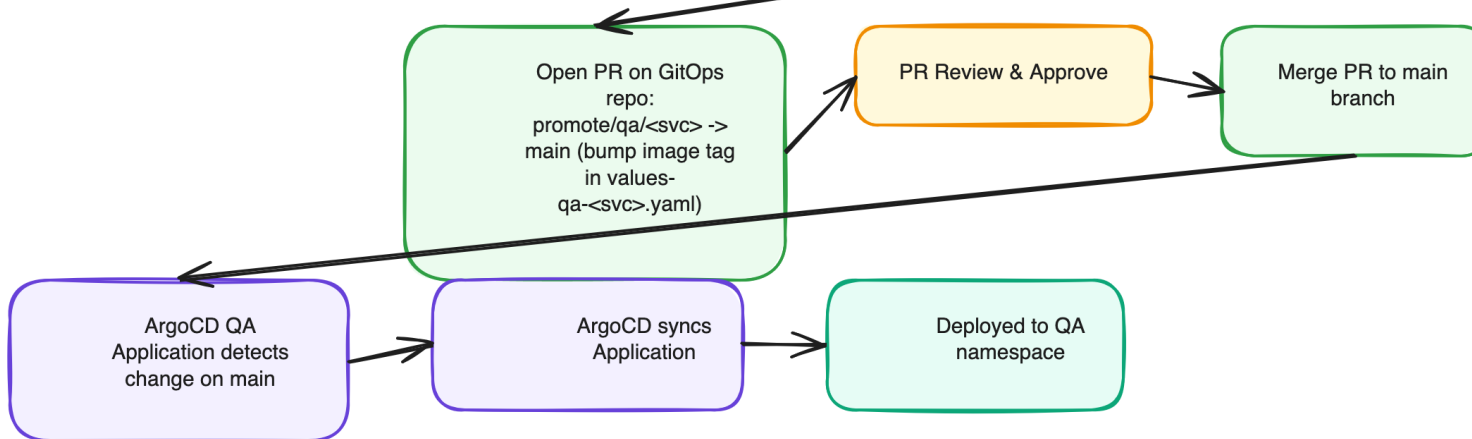
CI Pipeline: PR-check & Develop (ci.yaml)



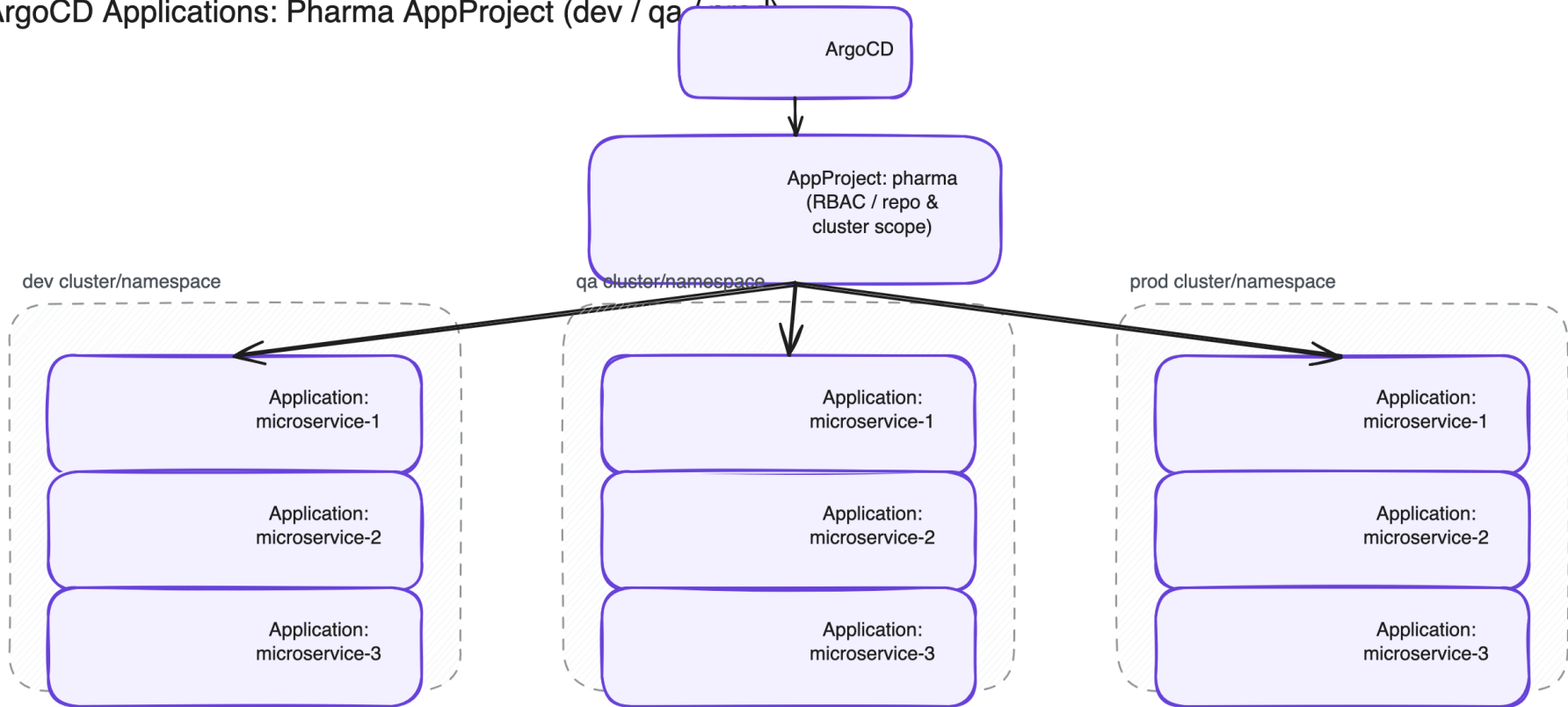
Dev CD : GitOps Deploy & Promotion Flow



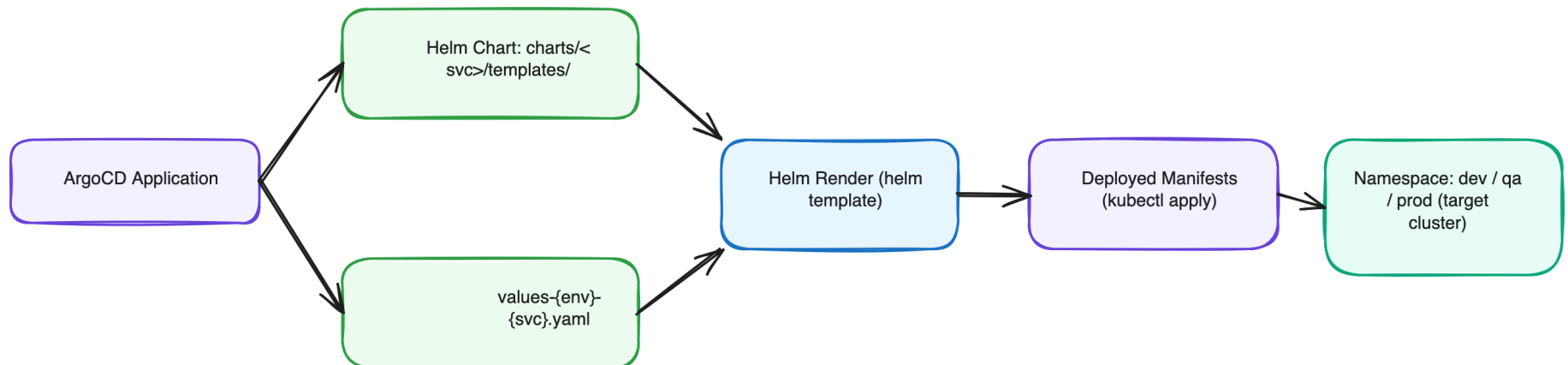
QA CD: GitOps Deploy & Promotion Flow



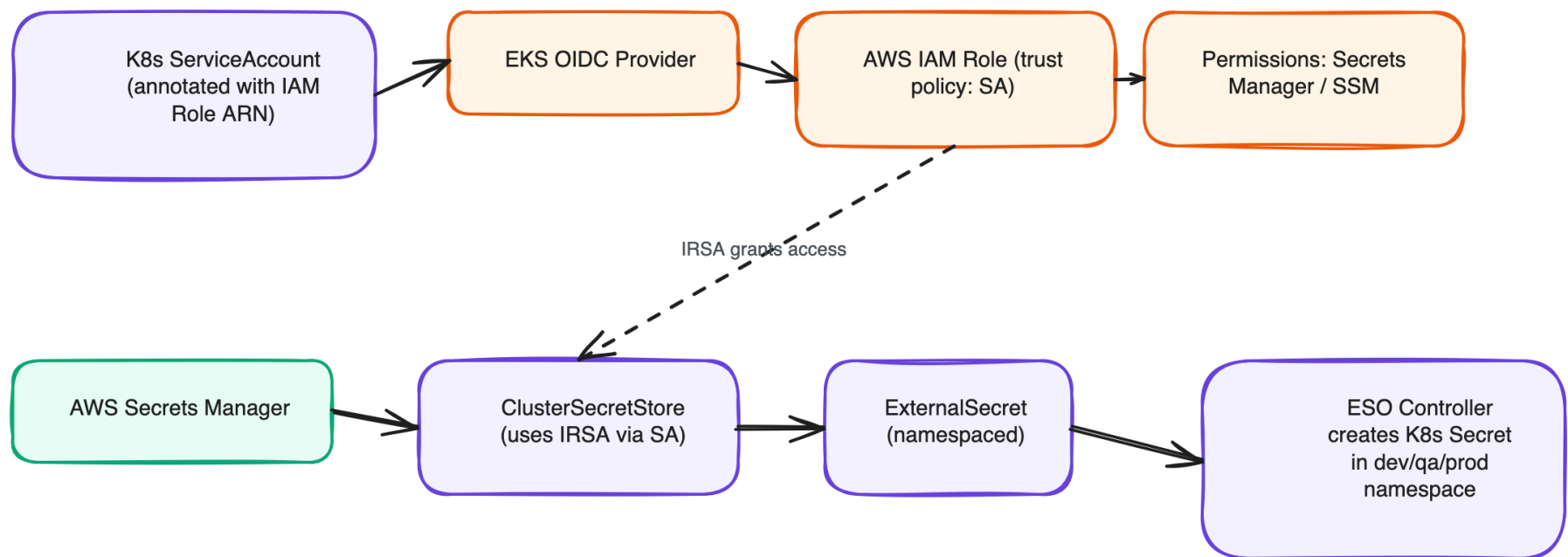
ArgoCD Applications: Pharma AppProject (dev / qa / prod)



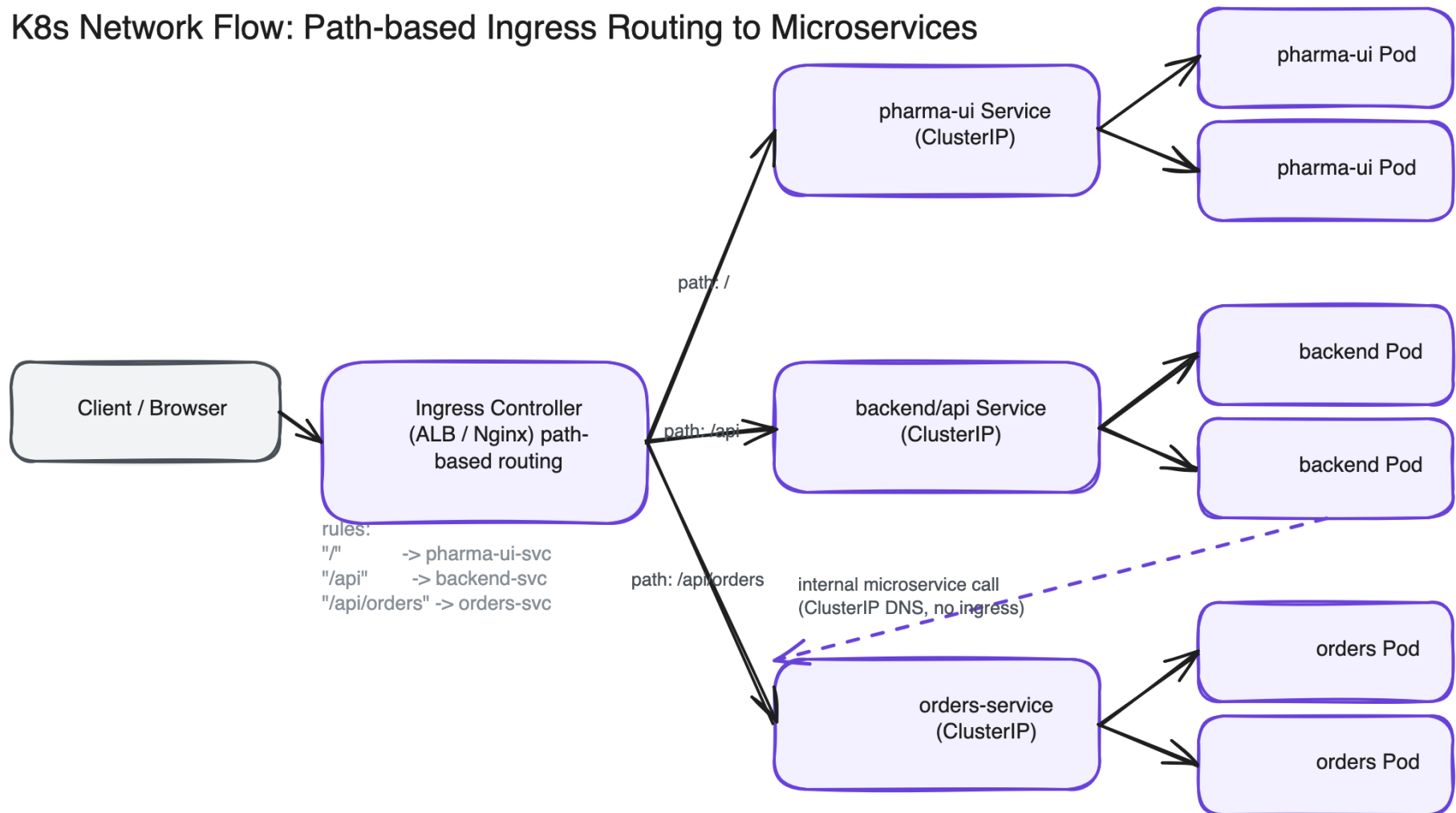
ArgoCD Application -> Helm Chart -> Namespace



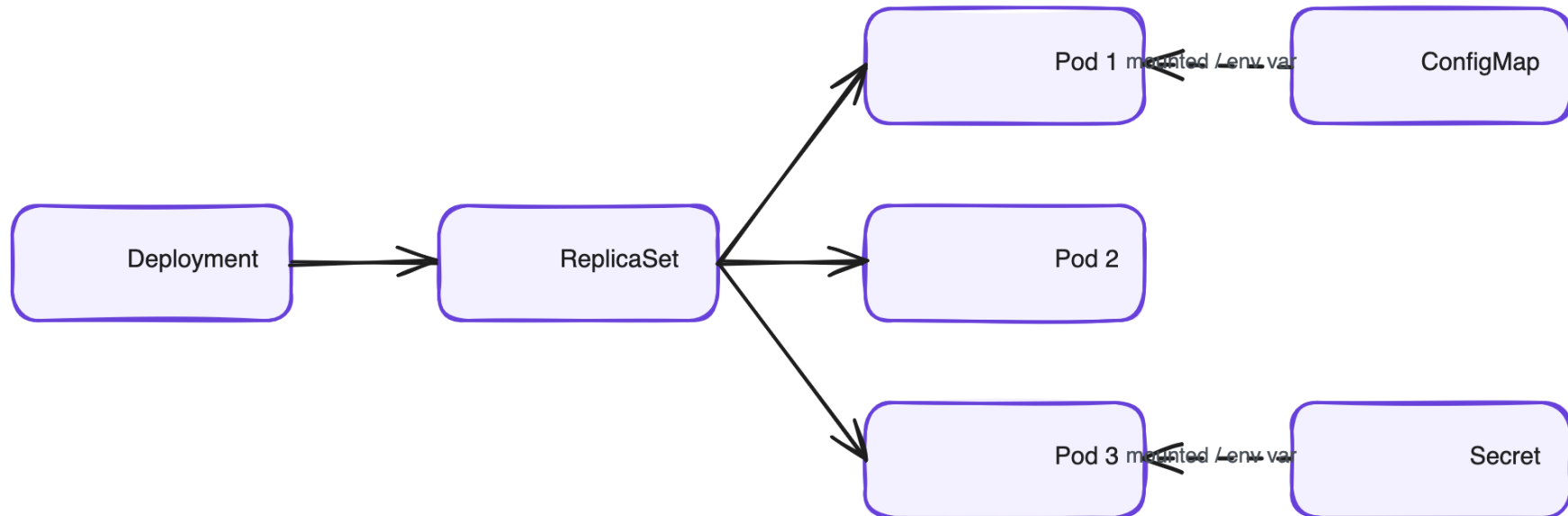
External Secrets Operator (ESO) + IRSA



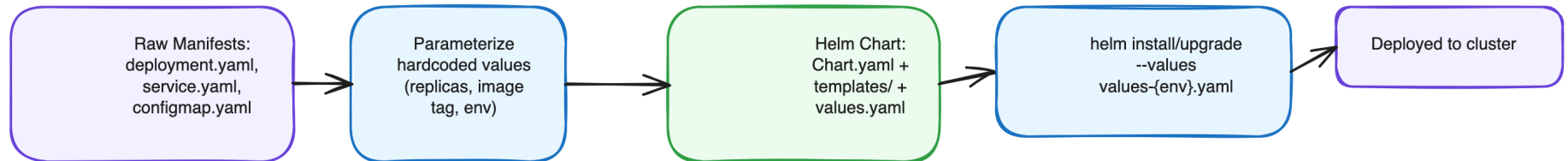
K8s Network Flow: Path-based Ingress Routing to Microservices

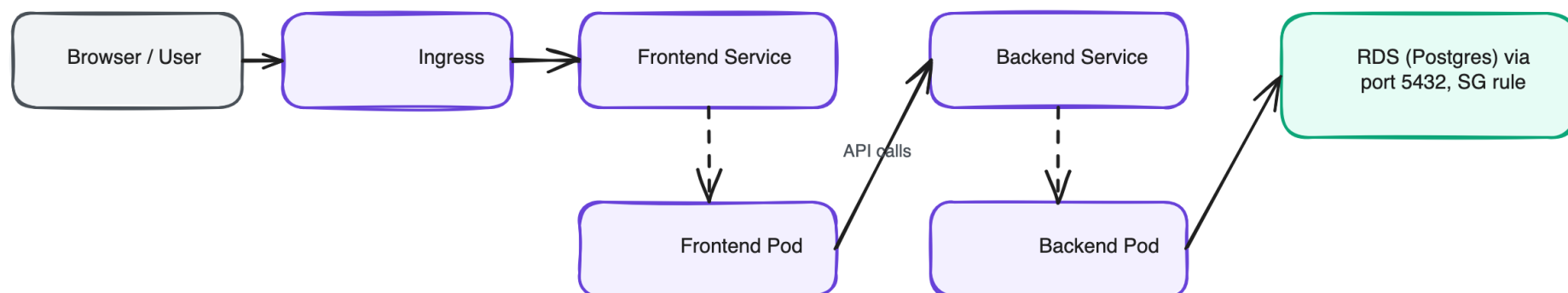


K8s Basics: Deployment -> ReplicaSet -> Pods



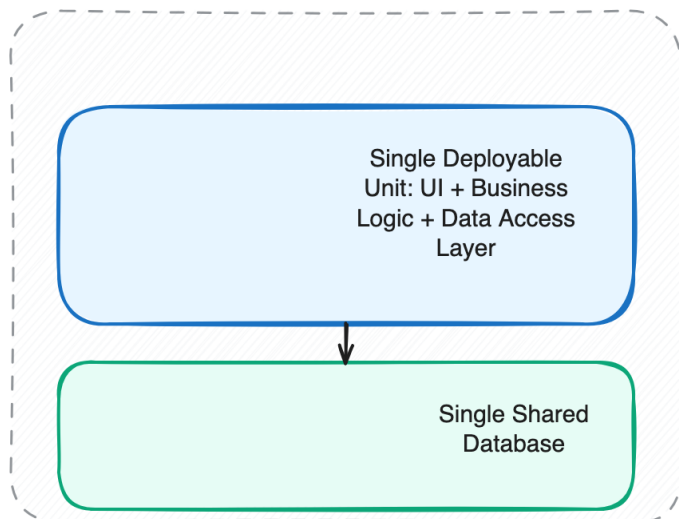
K8s Manifests -> Helm Chart (templates + values)



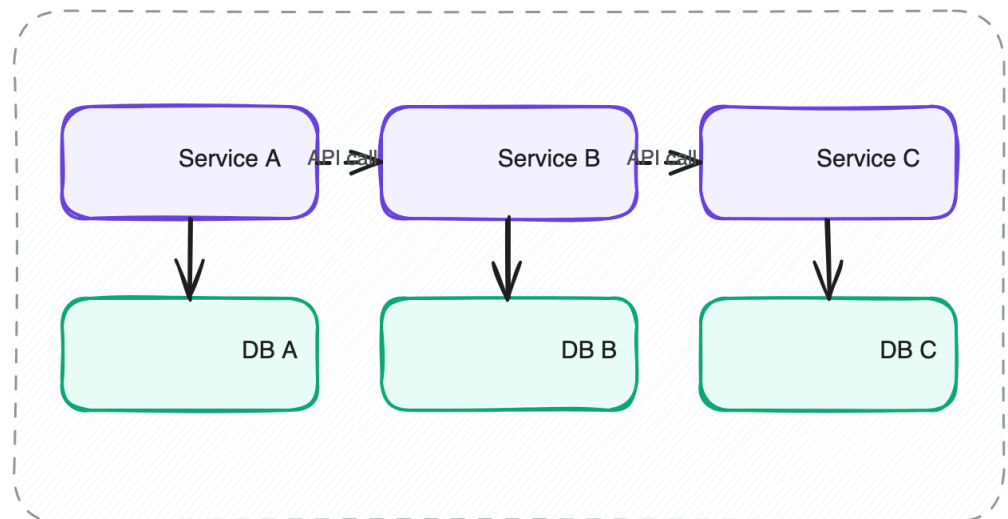
Network Flow: Frontend -> Backend -> RDS

Monolithic vs Microservices Architecture

Monolithic App

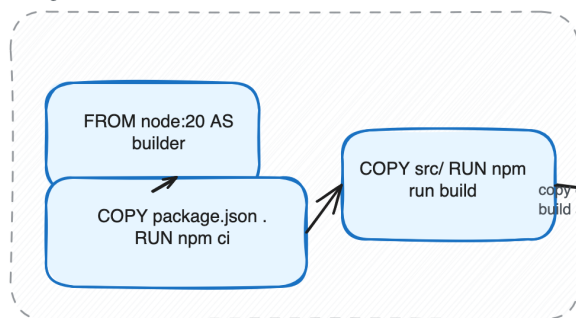


Microservices

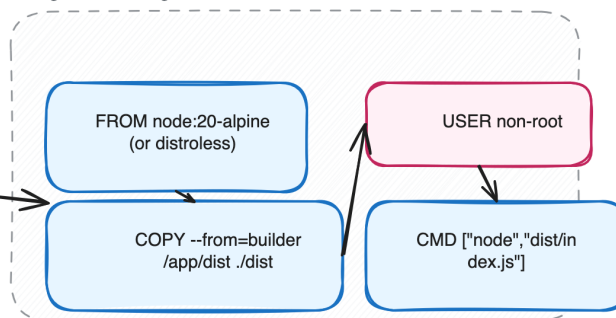


Dockerfile: Multi-stage Build & Optimization

Stage 1: builder



Stage 2: final image



Optimization tips: multi-stage build - small base image (alpine/distroless) - `.dockerignore`
- order `COPY` package.json before `src` for layer caching - non-root user - pin image versions